# Homework 2

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## Problem 1 Reformulate the following two problems as a linear program

### (a)

Rewrite to:

$$
\begin{aligned}
\min_{x \in \mathbb{R}^n} \quad & c^\top x + y \\
\text{s.t.} \quad & Ax \geq b, \\
& d^\top x = \alpha (\text{implicit}), \\
& y \geq \alpha, \\
& y \geq 0, \\
& y \geq 2\alpha - 4.
\end{aligned}
$$

### (b)

Rewrite to:

$$
\begin{aligned}
\min_{x_1, x_2, x_3} \quad & 2x_2 + t \\
\text{s.t.} \quad & t \geq x_1 - x_3, \\
& t \geq x_3 - x_1, \\
& x_1 + 2 \leq s, \\
& -(x_1 + 2) \leq s, \\
& x_2 \leq r, \\
& -x_2 \leq r, \\
& s + r \leq 5, \\
& x_3 \geq -1, \\
& x_3 \leq 1.
\end{aligned}
$$

## Problem 2 Consider the following linear programs

### (a)

Origin:

$$\begin{array}{ll} \max & 5x_1 + x_2 - 4x_3 \\ \text{s.t.} & x_1 + x_2 + x_3 + x_4 \geq 19 \\ & 4x_2 + 8x_4 \leq 45 \\ & x_1 + 6x_2 - x_3 = 7 \\ & x_1 \text{ unrestricted}, \quad x_2 \geq 0, \quad x_3 \geq 0, \quad x_4 \leq 0 \end{array}$$

Rewrite to:

$$\begin{array}{ll} \min & -5x_1^+ + 5x_1^- - x_2 + 4x_3 \\ \text{s.t.} & x_1^+ - x_1^- + x_2 + x_3 - x_7 - x_8 = 19 \\ & 4x_2 - 8x_7 + x_9 = 45 \\ & x_1^+ - x_1^- + 6x_2 - x_3 = 7 \\ & x_1^+, x_1^-, x_2, x_3, x_7, x_8, x_9 \geq 0 \end{array}$$

### b

Origin:

$$\begin{array}{ll} \min & 2x_1 - 7x_2 + 6x_3 + 5x_4 \\ \text{s.t.} & 2x_1 - 3x_2 - 5x_3 - 4x_4 \leq 20 \\ & 7x_1 + 2x_2 + 6x_3 - 2x_4 = 35 \\ & 4x_1 + 5x_2 - 3x_3 - 2x_4 \geq 15 \\ & 0 \leq x_1 \leq 10, 0 \leq x_2 \leq 8, x_3 \geq 2, x_4 \geq 0 \end{array}$$

Rewrite to:

$$\begin{array}{ll} \min & -2x_1 + 7x_2 + 6x_3 + 5x_4 \\ \text{s.t.} & -2x_1 + 3x_2 - 5x_3 - 4x_4 + x_5 = 34 \\ & 7x_1 + 2x_2 - 6x_3 + 2x_4 = 63 \\ & 4x_1 + 5x_2 + 3x_3 + 2x_4 + x_6 = 59 \\ & x_1, x_2, x_3, x_4, x_5, x_6 \geq 0 \end{array}$$

## Problem 3 Solve the following 3-dimensional linear optimization problem using the graphical method

**Active Constraints:** $x_1 + x_2 - x_3 = 0, x_1 = 4, x_1 + 2x_2 = 9$

**Optimal Solution:** $(x_1, x_2, x_3) = (4, 2.5, 6.5)$

**Feasible Region Vertices:** $(0,0), (0,2.5), (0.5,3), (3,3), (4,2.5), (4,0)$

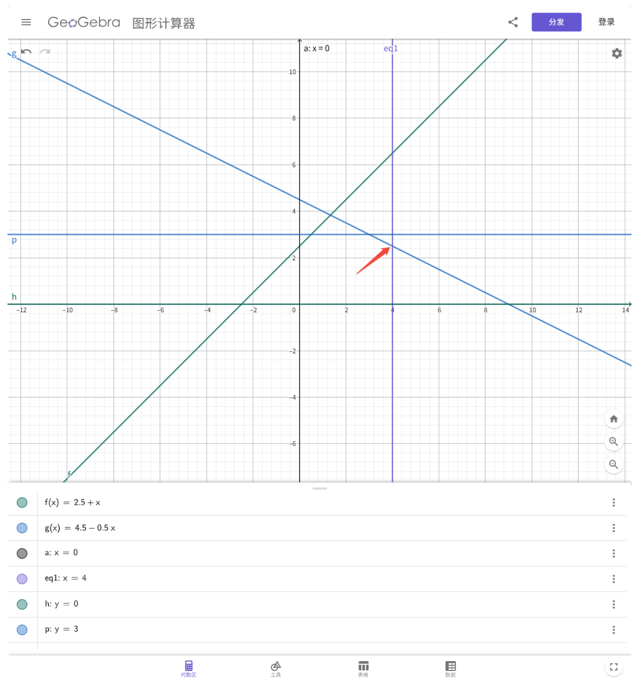


Figure 1: Graphical method

# Problem 4

First we can rewrite question in standard form:

$$
\begin{aligned}
\min \quad & -x_1 - x_2 \\
\text{s.t.} \quad & x_1 + 3x_2 - s_1 = 15, \\
& 2x_1 + x_2 - s_2 = 10, \\
& x_1 + 2x_2 + s_3 = 40, \\
& 3x_1 + x_2 + s_4 = 60, \\
& x_1, x_2, s_1, s_2, s_3, s_4 \geq 0.
\end{aligned}
$$

Then, consider with the first iteration, we have $x_1 = x_2 = 0$, which indicates that:

$$\begin{aligned}
x_1 &= 0 \\
x_2 &= 0 \\
s_1 &= 15 \\
s_2 &= 10 \\
s_3 &= 40 \\
s_4 &= 6
\end{aligned}$$

## Problem 5

### a

We have

$$\begin{array}{ll}
\min & y_1 + y_2 + y_3 \\
\text{s.t.} & x_1 - x_2 + 2x_3 - x_5 + y1 = 2, \\
& x_2 - x_3 + 2x_4 + x_6 + y_2 = 4, \\
& 2x_1 + 3x_3 - x_4 + y_3 = 2, \\
& 3x_1 + x_2 + s_4 = 60, \\
& x_i, y_i \geq 0.
\end{array}$$

### b

We have $x_1 = 2, x_4 = 2, y_2 = 0$,consider that $y_2 = 0$, it's not optimal.

## Problem 6

### a

False

$$\begin{array}{ll}
\min & x_3 \\
\text{s.t.} & x_1 - 3x_2 = 0, \\
& x_1, x_2, x_3 \geq 0.
\end{array}$$

We have $x_3 = 0$ as minimal, but $x_1, x_2$ unbounded.

### b

False

$$
\begin{array}{ll}
\min & x_1 + x_2 \\
\text{s.t.} & x_1 + x_2 = 1, \\
 & x_1, x_2 \geq 0.
\end{array}
$$

We have $x_1 = 0.5, x_2 = 0.5$, that is $m = 2 > 1$

### c

True

### d

False

Some of them are not bfs, but bs.